

Recommendations

1. **Air Filter to MAF Sensor**

Low Severity Leak

1. Inspect the air filter element for light dust loading or minor debris.
2. Verify the filter housing lid and seals are properly seated and not slightly open.
3. Continue the test and schedule a full filter inspection at the next planned maintenance.

Moderate Severity Leak

1. Inspect the air filter element for heavy dust or moisture intrusion.
2. Check the filter housing seal and intake bellows for cracks, damaged clips, or loose connections.
3. Verify the air filter service indicator (if present) and reset or replace the element as needed.
4. Pause testing after the current run and correct the leak before resuming full operation.

High Severity Leak

1. Stop the test immediately.
2. Replace the air filter element if it is contaminated or wet.
3. Repair or replace any cracked filter housing, intake bellows, or loose clamps.
4. Re-baseline the MAF (Mass Air Flow) and Delta-P sensor values before restarting the test.

2. **MAF Sensor to Turbocharger Compressor Inlet**

Low Severity Leak

1. Inspect intake hoses and clamps between the MAF sensor and turbo inlet for slight looseness.
2. Verify that the MAF sensor connections and mounting are secure (no small gaps or misalignment).
3. Continue the test while monitoring inlet pressure; plan to recheck hoses after the run.

Moderate Severity Leak

1. Tighten or replace any loose or damaged intake hose clamps and couplings.
2. Check for cracks or splits in the intake tubing and repair or replace as needed.
3. Ensure the MAF sensor is clean and properly calibrated (clean it if dusty).
4. Pause testing after the current run to resolve the leak and then recalibrate sensors before resuming.

High Severity Leak

1. Stop the test immediately.

2. Replace or repair the intake hose assembly between the MAF and turbo inlet.
3. Re-align and secure the MAF sensor and turbo inlet connections; ensure clamps are torqued.
4. After repair, re-baseline compressor inlet pressure and MAF readings before continuing.

3. **Compressor Outlet to Charge Air Cooler**

Low Severity Leak

1. Inspect the compressed-air hose and intercooler piping for minor looseness or wear.
2. Check that the compressor outlet to CAC connection flange is tight and free of small gaps.
3. Continue testing with monitoring; schedule a thorough check of intercooler hoses at the next break.

Moderate Severity Leak

1. Tighten or replace loose hose clamps and couplings on the intercooler charge pipe.
2. Inspect the charge-air cooler core and connections for leaks; repair any cracks or holes.
3. Verify sensor readings on the pressure/temperature sensor and recalibrate if they're inconsistent.
4. Pause testing after this run to fix the leak before proceeding.

High Severity Leak

1. Stop the test immediately.
2. Replace or repair any damaged intercooler piping or compressor outlet housings.
3. Ensure all clamps, couplings, and seals between the compressor and CAC are fully intact.
4. Re-baseline compressor outlet pressure and temperature readings before restarting the engine.

4. **Charge Air Cooler (CAC) to Intake Manifold**

Low Severity Leak

1. Inspect intake manifold hoses and couplings for small leaks or loose clamps.
2. Check that the MAP and MAT sensor connections are tight and the sensors are clean.
3. Continue the test, but log any slight drop in manifold pressure for follow-up.

Moderate Severity Leak

1. Tighten or replace worn clamps, gaskets, or hoses between the CAC outlet and intake manifold.
2. Inspect the CAC outlet flange and intake manifold gasket for leaks; reseal or replace as needed.

3. Verify throttle body and intake butterfly seals if present, ensuring no bypass air.
4. Plan to stop after this test run to perform repairs before continuing full-load tests.

High Severity Leak

1. Stop the test immediately.
2. Repair or replace any faulty intake hoses, clamps, or manifold gaskets.
3. Ensure all bolts on the intake manifold are properly torqued.
4. Re-calibrate the MAP/MAT sensor and re-baseline intake pressure readings before resuming.

5. Cylinder to Turbocharger Turbine Inlet

Low Severity Leak

1. Inspect exhaust manifold bolts and turbo inlet flange for minor looseness.
2. Look for small hairline cracks at manifold joints or turbine inlet.
3. Continue testing with caution; monitor turbine inlet pressure and exhaust sounds.

Moderate Severity Leak

1. Tighten exhaust manifold flange bolts and replace any damaged gaskets.
2. Inspect and repair any cracks or gaps in the exhaust manifold or turbo inlet pipe.
3. Check the turbocharger turbine housing for leaks or loose mounting studs; reseal as needed.
4. Schedule a test shutdown after the current run to address and seal the leak.

High Severity Leak

1. Stop the test immediately.
2. Repair or replace the exhaust manifold or turbine inlet components (gaskets, flanges, studs).
3. Ensure the turbocharger turbine housing is tightly secured and free of cracks.
4. After repair, re-baseline turbine inlet pressure and recalibrate related engine controls before continuing.

6. Diesel Oxidation Catalyst

Low Severity Leak

1. Inspect the DOC inlet flange and gaskets for small leaks.
2. Check that the DOC mounting bracket and bolts are secure and not slightly loose.

3. Continue the test with logging of EGT inlet/outlet; plan inspection during next maintenance.

Moderate Severity Leak

1. Tighten or replace DOC inlet flange bolts and gaskets.
2. Inspect the exhaust manifold and turbo outlet for leaks that could affect DOC flow.
3. Check the DOC mounting structure for cracks or deformation; repair if needed.
4. Pause the test after this run to repair the leak and prevent emissions violations.

High Severity Leak

1. Stop the test immediately.
2. Replace any cracked or damaged DOC inlet pipes, flanges, or gaskets.
3. Ensure the DOC is properly seated and secured; check for catalyst damage.
4. After repair, re-baseline EGT sensors and confirm DOC temperature readings before resuming.

7. Diesel Particulate Filter

Low Severity Leak

1. Inspect the DPF housing and inlet for loose clamps or tiny cracks.
2. Check the differential pressure sensor connections for secure fit and cleanliness.
3. Continue testing; schedule a closer check of the DPF seal during the next idle or shutdown.

Moderate Severity Leak

1. Tighten or replace any leaking clamps, seals, or pipe sections upstream of the DPF.
2. Inspect the DPF inlet for any filter bypass paths or damaged carrier surfaces.
3. Verify proper operation of the DPF regeneration cycle; forced regeneration may be required after repair.
4. Pause testing to fix the leak and prevent particulate emissions issues.

High Severity Leak

1. Stop the test immediately.
2. Repair or replace the DPF assembly including any damaged pipes or seals.
3. Re-calibrate the DPF differential pressure sensor after fixing the leak.
4. After repair, perform a baseline emissions check before resuming tests.

8. **Selective Catalytic Reduction**

Low Severity Leak

1. Inspect SCR inlet piping and injector connections for slight looseness.
2. Verify urea tank level and injection system operation.
3. Continue testing with NOx monitoring; check again after the run if values creep up.

Moderate Severity Leak:

1. Tighten or replace leaking clamps or seals on the SCR inlet piping.
2. Check for any clogged or malfunctioning urea injectors and clean/replace if needed.
3. Ensure the SCR catalyst is properly engaged and not bypassed.
4. Schedule a stop after this run to repair the leak and recalibrate NOx sensors.

High Severity Leak:

1. Stop the test immediately.
2. Repair or replace any damaged SCR pipes, clamps, or injection system components.
3. Flush and recharge the urea delivery system if contaminated.
4. After repair, re-baseline the NOx sensor readings and confirm SCR operation before continuing.

9. **No Leak Detected-Healthy**

1. Baseline performance confirmed.
2. Proceed to next test stage or clear engine for shipment.
3. No mechanical intervention required.